

James Hobin

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github.com/hobinjk hobinjk.github.io/resume

EDUCATION

Massachusetts Institute of Technology 2012–2017

- Obtained Master of Engineering in Computer Science from the MIT Media Lab in 2017
- Obtained Bachelor of Science in Computer Science in 2016

EXPERIENCE

PTC Inc. distributed systems engineering augmented reality industrial iot artificial intelligence 2019–2024

Senior Principal Innovation Engineer

- Created a machine-learning-enabled tool which cuts the time for factory work analysis by 80%
- Developed a container-based cloud host for real-time multi-user contextual XR experiences
- Invented depth streaming protocol with peer-to-peer connections and storage format
- Led testing and developer tooling efforts across all Reality Lab projects
- Granted two patents for original research (US11776205B2, US11625140B2)
- 2nd author on CHI 2024 publication "The Pop-Up Metaverse: A Multi-User, Multi-Tasking Spatial Computing Environment for Collaborative Spatial Problem-Solving"

Mozilla javascript node linux embedded devices c 2017–2019

Platform Engineer, Advanced Development

- Developed a C library for any Arduino-compatible device to expose the Web of Things API
- Created a comprehensive rules engine for automating smart home functionality
- Designed, implemented, and tested the Mozilla IoT Gateway's OAuth security framework
- Wrote a series of technical blog posts for Mozilla Hacks (90,000+ followers)

MIT Media Lab c++ ios javascript node 2012–2017

Fluid Interfaces Researcher

- Designed adaptable, modular interfaces for the Reality Editor, a novel AR IoT platform
- Refined inter-device protocols for reliability and speed
- 2nd author on UBICOMP 2013 publication "Reality Editor: Programming Smarter Objects"

Mozilla c++ javascript android internals 2014, 2015

Firefox OS Systems Front End Team Intern

- Produced several major features and improvements over two summer internships
- Spearheaded research project to uncover a significant market opportunity for Firefox OS

PROJECTS

New XKit javascript browser extension open source 2015–2019

- Forked and revitalized a popular open source project for improving Tumblr
- Supported over 400,000 users across Chrome, Firefox, and Safari

Art and Games javascript c++ rust writing webgl 2010–Present

- League of Lines: Voice-line-based online competitive card game
- Full portfolio: <https://hobinjk.github.io/portfolio>

Cabbage Legs javascript machine learning canvas 2015

- Programmed a genetic algorithm that optimizes simulated quadrupedal robots
- Extended into an online game for crowdsourcing the design of robots as undergraduate researcher at CSAIL's Distributed Robotics Laboratory